HireHub – A Full Stack MERN-Based Job Portal for Students and Recruiters

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***Abstract*—HireHub is a full stack web-based job portal developed using the MERN stack, namely MongoDB, Express.js, React.js, and Node.js, to simplify and modernize the recruitment process for students and recruiters. The system provides a common digital platform where students can create profiles, explore job opportunities, and apply for suitable positions, while recruiters can post jobs, manage applicants, and monitor recruitment activities efficiently. The platform includes secure JWT-based authentication, role-based access control, job and application management, recruiter approval by admin, and responsive user interfaces. HireHub aims to reduce the communication gap between students and employers and provide a scalable, secure, and user-friendly recruitment solution.**

***Index Terms*—MERN Stack, Job Portal, Recruitment System, React.js, Node.js, MongoDB, Express.js, JWT Authentication, Role-Based Access Control**

I. Introduction

In the digital era, online recruitment platforms have become essential for connecting job seekers with recruiters in an efficient and organized manner. Traditional recruitment methods often involve manual communication, scattered job postings, delayed application tracking, and limited transparency for both students and employers. To address these issues, a smart and centralized recruitment system is required.

HireHub is a web-based recruitment platform designed to bridge the gap between students seeking employment opportunities and recruiters searching for suitable candidates. The platform is built using the MERN stack, which supports the development of modern, scalable, and responsive full stack web applications. HireHub allows students to register, browse jobs, apply for positions, and manage their applications, while recruiters can post jobs, view applicants, and update application statuses. The platform also includes an admin module for managing users, jobs, and recruiter approvals.

The project focuses on creating a secure, user-friendly, and role-based job portal that improves accessibility, transparency, and efficiency in the hiring process.

II. Problem Statement

Many students face difficulty in finding relevant job opportunities through a single reliable platform, while recruiters often struggle to manage applications efficiently using traditional methods. Existing recruitment methods may lack proper filtering, centralized management, role-based access, and real-time transparency. Hence, there is a need for a modern job portal that simplifies job posting, job searching, application submission, and recruiter-candidate interaction through a secure and structured web application.

III. Objectives of the Study

The main objectives of the HireHub project are as follows:

- To design and develop a web-based job portal for students and recruiters.
- To provide secure user authentication and role-based access control.
- To enable recruiters to post job opportunities and manage applicants.
- To allow students to search and apply for relevant job openings easily.
- To build an admin module for managing users, jobs, and recruiter approval.
- To create a responsive, modern, and user-friendly interface.
- To improve efficiency, transparency, and accessibility in the recruitment process.

IV. Literature Review

Several online job portals and recruitment systems already exist in the market, such as LinkedIn, Naukri, Indeed, and Internshala. These platforms provide useful services for employers and job seekers, including profile building, job search, and online applications. However, many existing systems are either too broad, overly complex, or not specifically tailored to the academic and early-career needs of students.

Research on web-based recruitment systems shows that modern full stack technologies improve accessibility, usability, and scalability. MERN stack applications are especially popular because they allow developers to build complete web solutions using JavaScript across both frontend and backend. Studies also suggest that secure authentication, centralized job management, and role-based access are important factors in creating effective recruitment platforms.

Based on this review, HireHub is proposed as a focused and practical recruitment platform that combines the strengths of

modern web technology with student-oriented job access and recruiter management features.

V. Proposed System / Methodology

The proposed system is HireHub, a full stack web application developed using MongoDB, Express.js, React.js, and Node.js. The development followed a modular and structured approach.

The backend was developed first to handle authentication, job posting, application processing, and role-based authorization through RESTful APIs. MongoDB was used for storing user, job, and application data. JWT authentication was implemented to secure routes and sessions.

The frontend was then developed using React.js with Tailwind CSS to create a responsive and interactive user interface. Different pages and components were designed for students, recruiters, and admin users. Axios was used for communication between frontend and backend services.

The overall methodology involved requirement analysis, system design, database design, API development, frontend implementation, integration, testing, debugging, and deployment preparation.

VI. System Design

A. Architecture Diagram

The architecture of HireHub follows a three-tier structure:

1) **Presentation Layer:** React.js frontend with Tailwind CSS for UI.
2) **Application Layer:** Node.js and Express.js backend handling APIs and business logic.
3) **Data Layer:** MongoDB database for storing users, jobs, and applications.

B. UML / Flow Description

The primary actors in the system are:

- **Student:** Registers, logs in, views jobs, applies for jobs, checks application status.
- **Recruiter (HR):** Registers, logs in after approval, posts jobs, manages applicants, updates application status.
- **Admin:** Manages users, jobs, and recruiter approval.

The main workflow is:

1) User registers and logs into the system.
2) Student searches available jobs and submits applications.
3) Recruiter posts jobs and reviews applicants.
4) Recruiter updates applicant status as pending, selected, or rejected.
5) Admin approves or rejects recruiter accounts and monitors the platform.

C. Database Design

The main collections used in MongoDB are `Users`, `Jobs`, and `Applications`.

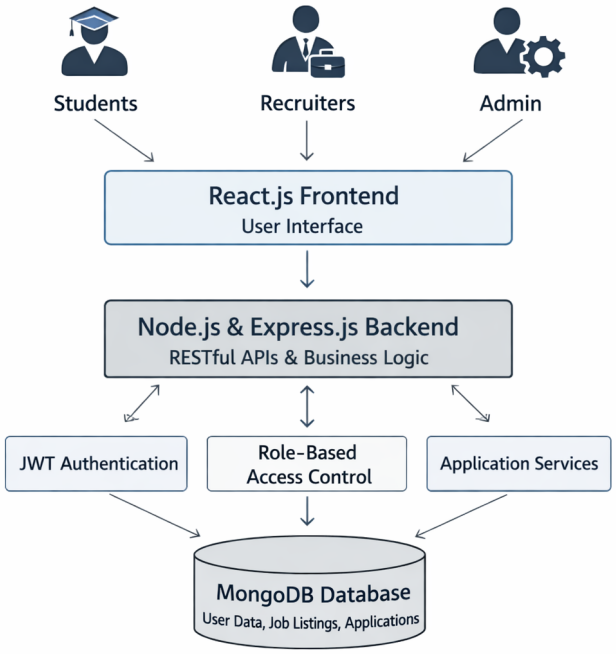

Fig. 1. System Architecture of HireHub.

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1) User Collection:

- `_id`
- `name`
- `email`
- `password`
- `role` (student / hr / admin)
- `isApproved`
- `resume`
- `bio`
- `skills`

2) Job Collection:

- `_id`
- `title`
- `company`
- `location`
- `salary`
- `description`
- `postedBy`

3) Application Collection:

- `_id`
- `job`

- `student`
- `status`
- `appliedAt`

VII. Implementation

HireHub was implemented using a full stack JavaScript approach. The backend was developed with Node.js and Express.js, where REST APIs were created for user registration, login, job posting, job retrieval, application submission, and applicant status management. JWT was used for authentication and middleware was added for protected routes and role-based access.

The frontend was developed in React.js using reusable components and routing. Tailwind CSS was used for styling and responsiveness. Separate pages were created for Home, Login, Register, Jobs, Post Job, Applicants, My Applications, Recruiter Dashboard, and Admin Panel. Axios was used for API integration and local storage was used for managing authentication state.

Major implemented features include:

- User registration and login with JWT authentication
- Role-based system for student, recruiter, and admin
- Recruiter approval system controlled by admin
- Job posting, updating, and deletion by recruiter
- Job browsing and application system for students
- Applicant tracking and status updates
- Admin management of users and jobs
- Responsive UI with dark/light mode support

VIII. Results and Analysis

The developed HireHub system successfully achieved the intended functionality of a job portal for students and recruiters. Testing showed that users could register and log in securely, recruiters could post jobs, students could apply for positions, and admins could manage recruiter approvals and platform data.

The system improved the recruitment workflow by centralizing job information and applicant data in one platform. Students were able to track applications more easily, while recruiters could review candidates in a structured manner. The responsive user interface and role-based navigation also improved usability across different types of users.

The MERN stack architecture proved effective for building a scalable and maintainable application. MongoDB supported flexible document storage, Express.js and Node.js handled server-side operations efficiently, and React.js enabled interactive frontend rendering.

IX. Discussion

The HireHub project demonstrates how modern web technologies can be used to solve real-world recruitment challenges in an academic and practical context. By combining secure authentication, role-based dashboards, job management, and applicant tracking, the system offers a complete recruitment workflow within a single platform.

One important strength of the project is its modular architecture, which allows future expansion. Additional features such as resume parsing, interview scheduling, recommendation systems, and analytics dashboards can be integrated without redesigning the whole platform. However, some advanced features such as AI-based job matching, email automation, and large-scale deployment optimization are still outside the current scope.

Overall, the project serves as a strong example of applying full stack development principles to create a useful and industry-relevant software solution.

X. Conclusion

HireHub is a practical and efficient full stack job portal designed to connect students and recruiters through a centralized digital platform. The project successfully implements core recruitment functionalities including registration, login, job posting, job application, applicant management, and admin control. By using the MERN stack, the platform achieves flexibility, responsiveness, and scalability. The system reduces manual effort, improves transparency, and provides a better recruitment experience for both job seekers and employers. Therefore, HireHub can be considered a valuable academic project with strong real-world relevance.

XI. Future Scope

The future scope of HireHub includes several enhancements that can improve its functionality and user experience:

- Integration of resume upload and parsing features
- Email notifications for job applications and status updates
- AI-based job recommendation system
- Interview scheduling and meeting integration
- Admin analytics dashboard with charts and reports
- Advanced job filtering and search optimization
- Cloud deployment and production-level security enhancements

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References

[1] MongoDB Documentation, "MongoDB Manual," [Online]. Available: https://www.mongodb.com/docs/

[2] React Documentation, "React Official Documentation," [Online]. Available: https://react.dev/

[3] Node.js Documentation, "Node.js Official Documentation," [Online]. Available: https://nodejs.org/en/docs

[4] Express.js Documentation, "Express – Node.js Web Application Framework," [Online]. Available: https://expressjs.com/

[5] Mozilla Developer Network, "JSON Web Tokens," [Online]. Available: https://developer.mozilla.org/

[6] T. Berners-Lee, R. Fielding, and H. Frystyk, "Hypertext Transfer Protocol – HTTP/1.0," RFC 1945, May 1996.

[7] M. Fowler, *Patterns of Enterprise Application Architecture*. Boston, MA, USA: Addison-Wesley, 2002.

[8] I. Sommerville, *Software Engineering*, 10th ed. Boston, MA, USA: Pearson, 2015.

Appendix A
Sample API Endpoints

Some sample API endpoints used in the system are listed below:

- `POST /api/auth/register`
- `POST /api/auth/login`
- `GET /api/jobs`
- `POST /api/jobs`
- `POST /api/applications`
- `GET /api/applications/my`
- `PUT /api/applications/status/:applicationId`

Appendix B
Modules of the System

The system is divided into the following modules:

- Student Module
- Recruiter Module
- Admin Module
- Authentication Module
- Job Management Module
- Application Management Module